

DNS Log Analysis Using Splunk Project

By Pranali Sonar

Project Overview

This project demonstrates how to ingest, parse, and analyze DNS logs using Splunk Enterprise.

By leveraging Zeek-style JSON DNS logs, we perform meaningful security and traffic analysis using Splunk Search Processing Language (SPL).

The lab focuses on identifying:

- Frequently queried domain names
- Most active client IPs
- Distribution of DNS record types
- Potential anomalies in DNS behaviour

Objectives

By completing this project, you will learn how to:

- Ingest JSON-formatted DNS logs into Splunk
- Extract useful DNS metadata (queries, IPs, record types)
- Write SPL queries for DNS traffic analysis
- Detect suspicious or abnormal DNS activity
- Build a strong foundation for DNS-based threat detection

Tools & Technologies

- Splunk Enterprise
 - Zeek DNS Logs (JSON format)
 - Search Processing Language (SPL)
-

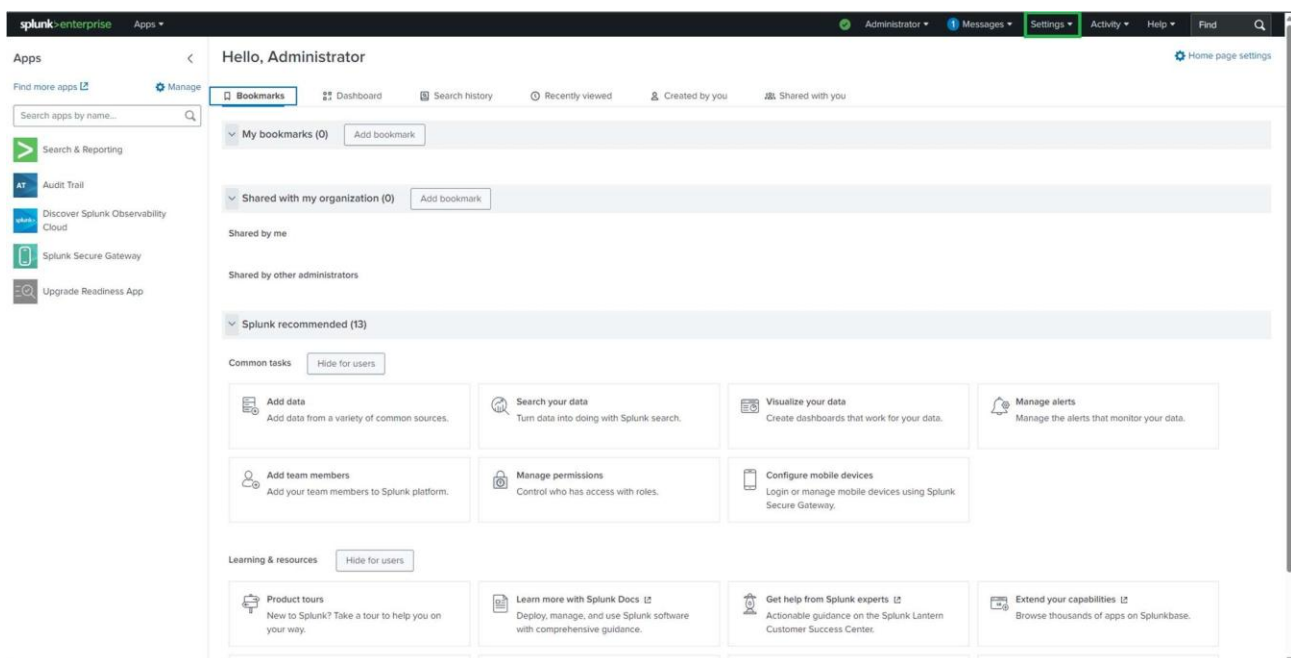
Dataset Information

The dataset consists of Zeek-style DNS logs in JSON format with fields such as: ts
→ Timestamp id.orig_h → Source IP address id.resp_h → DNS server IP qtype
→ DNS query type (A, AAAA, PTR, CNAME) query → Queried domain name
answers → DNS response rcode → DNS response code rtt → Round Trip
Time

Lab Setup & Data Ingestion

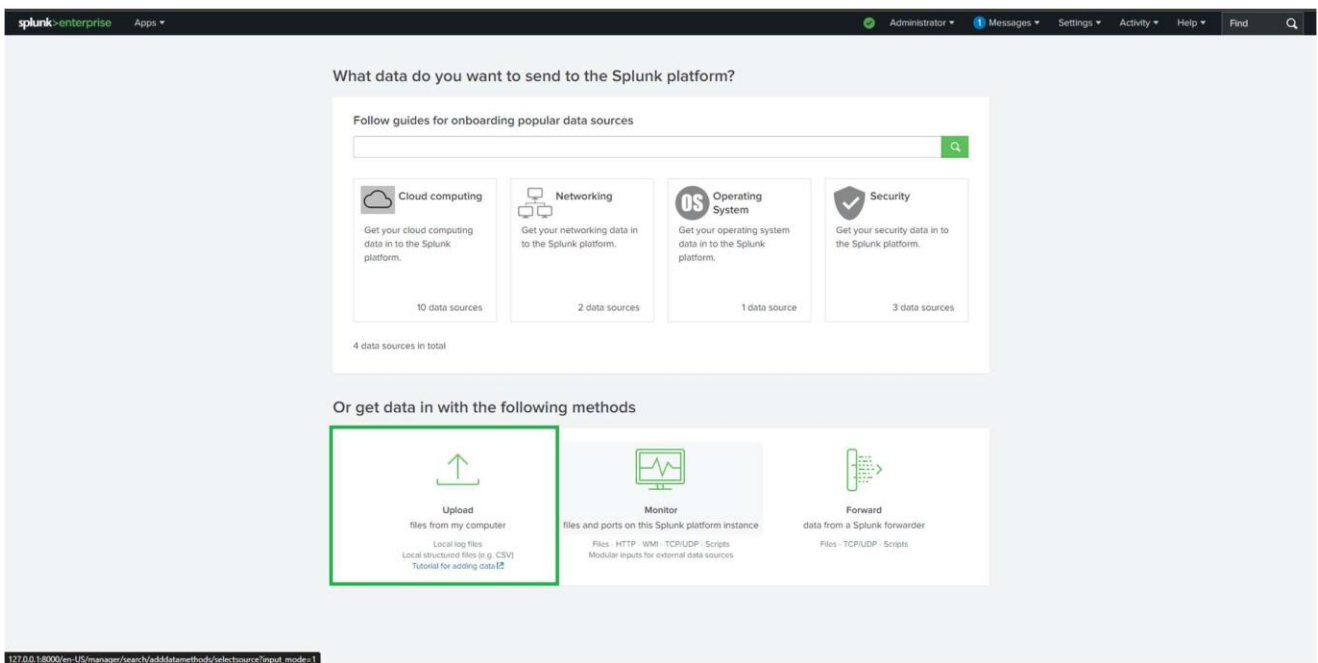
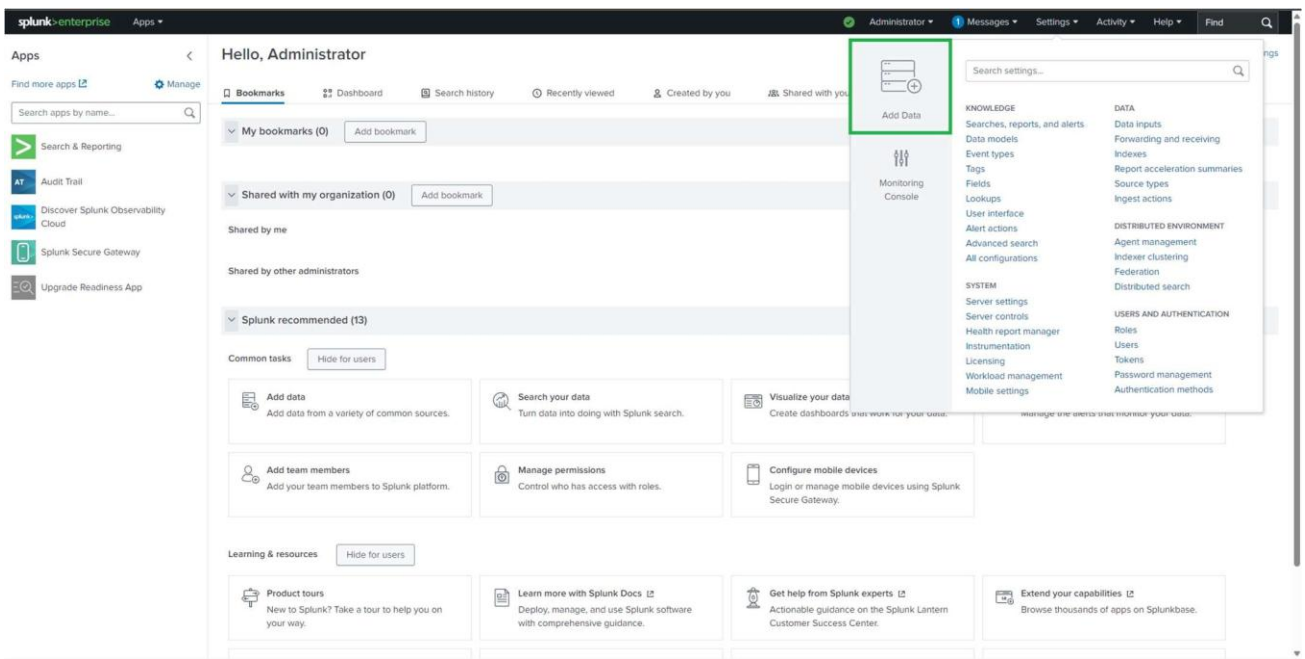
Step 1: Upload DNS Logs

1. Open Splunk Web

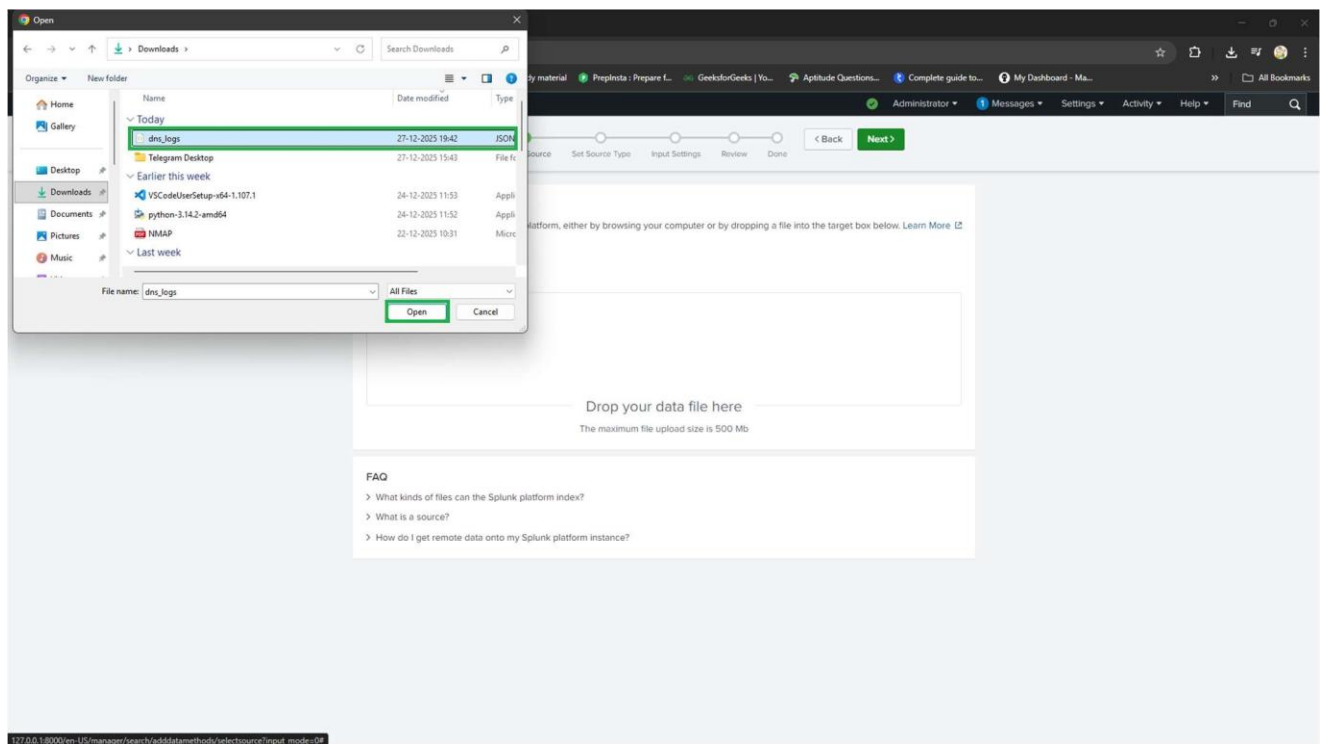
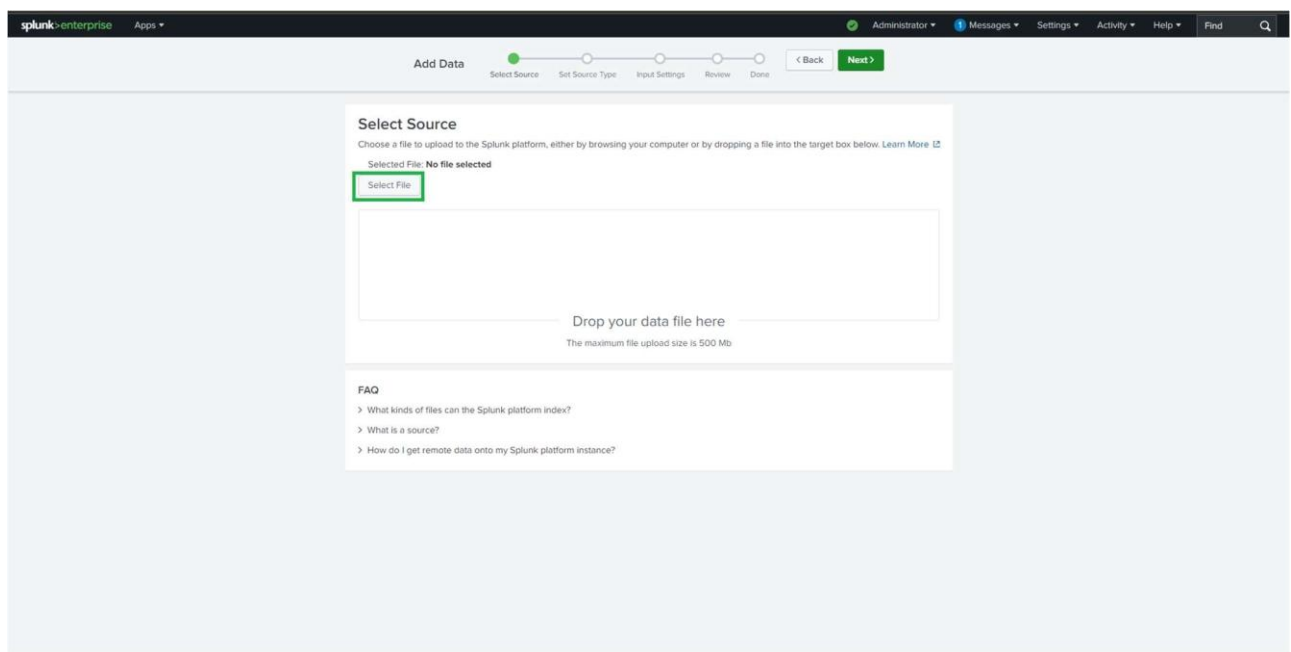


2. Navigate to:

Settings → Add Data → Upload



3. Select the file:



4. dns_logs.json

splunk>enterprise Apps ▾

Administrator ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find 🔍

Add Data Select Source Set Source Type Input Settings Review Done < Back Next >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: **dns_logs.json**

Select File

Drop your data file here

The maximum file upload size is 500 Mb

File Successfully Uploaded

FAQ

- > What kinds of files can the Splunk platform index?
- > What is a source?
- > How do I get remote data onto my Splunk platform instance?

5. Set the following:

- Source Type: json (or custom zeek:dns)

Index: dns_lab (recommended)

splunk>enterprise Apps ▾

Administrator ▾ Messages ▾ Settings ▾ Activity ▾ Help ▾ Find 🔍

Add Data Select Source Set Source Type Input Settings Review Done < Back Next >

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **dns_logs.json**

Source type: **json** Save As

Format Select Select

View Event Summary

	Time	aa	answers	id.orig_h	id.orig_p	id.resp_h	id.resp_p	proto	qclass	qtype	query	ra	rcode	rd	rejected	rt
1	4/23/25 2:51:09.069 PM	false	alias.google.com	192.168.112	58167	192.168.11	53	udp	IN	CNAME	google.com	true	NOERROR	true	false	0.397165
2	4/23/25 2:51:09.069 PM	false	192.168.10134	192.168.112	13629	192.168.13	53	udp	IN	A	fileservr.local	true	NOERROR	true	false	0.451793
3	4/23/25 2:51:09.069 PM	false	192.168.40347	192.168.118	40231	192.168.13	53	udp	IN	A	backup.local	true	NOERROR	true	false	0.458119
4	4/23/25 2:51:09.069 PM	false	alias.fileservr.local	192.168.117	3446	192.168.12	53	udp	IN	CNAME	fileservr.local	true	NOERROR	true	false	0.23946
5	4/23/25 2:51:09.069 PM	false	192.168.117in-addr.arpa	192.168.117	25615	192.168.12	53	udp	IN	PTR	google.com	true	NOERROR	true	false	0.30075
6	4/23/25 2:51:09.069 PM	false	192.168.117in-addr.arpa	192.168.117	16271	192.168.12	53	udp	IN	PTR	ip6test.local	true	NOERROR	true	false	0.13411
7	4/23/25 2:51:09.069 PM	false	192.168.208.219	192.168.116	63770	192.168.12	53	udp	IN	A	google.com	true	NOERROR	true	false	0.091741
8	4/23/25 2:51:09.069 PM	false	192.168.195.9	192.168.110	19707	192.168.13	53	udp	IN	A	microsoft.com	true	NOERROR	true	false	0.167656
9	4/23/25 2:51:09.069 PM	false	alias.backup.local	192.168.114	32719	192.168.13	53	udp	IN	CNAME	backup.local	true	NOERROR	true	false	0.34248
10	4/23/25 2:51:09.069 PM	false	192.168.82.34	192.168.112	8592	192.168.11	53	udp	IN	A	router.local	true	NOERROR	true	false	0.44890
11	4/23/25 2:51:09.069 PM	false	alias.printer.local	192.168.117	29930	192.168.12	53	udp	IN	CNAME	printer.local	true	NOERROR	true	false	0.31964
12	4/23/25 2:51:09.069 PM	false	192.168.70.240	192.168.121	57366	192.168.13	53	udp	IN	A	google.com	true	NOERROR	true	false	0.411965

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AdministratorMessagesSettingsActivityHelpFind

Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< BackReview >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

☒ Constant value

☐ Regular expression on path

☐ Segment in path

Host field valueLAPTOP-EHJ3QFJI

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

IndexDefaultCreate a new index

FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

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New Index

General Settings

Index Name

Set index name (e.g., INDEX_NAME). Search using index=INDEX_NAME.

Index Data Type

EventsMetrics

The type of data to store (event-based or metrics).

Home Path

optional

Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold Path

optional

Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb).

Thawed Path

optional

Thawed/restricted db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb).

Data Integrity Check

EnableDisable

Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index

500GB

Maximum target size of entire index.

Max Size of Hot/Warm/Cold Bucket

autoGB

Maximum target size of buckets. Enter 'auto_high_volume' for high-volume indexes.

Frozen Path

optional

Frozen bucket archive path. Set this if you want Splunk to automatically archive frozen buckets.

App

Search & Reporting

Storage Optimization

SaveCancel

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Help

Find

New Index

General Settings

Index Name

dns_lab

Set index name (e.g., INDEX_NAME). Search using index=INDEX_NAME.

Index Data Type

Events

The type of data to store (event based or metrics).

Home Path

optional

Hotwarm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/idx).

Cold Path

optional

Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/cold).

Thawed Path

optional

Thawed/unfrozen db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thawed).

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Enable

Disable

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Max Size of Entire Index

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Maximum target size of entire index.

Max Size of Hot/Warm/Cold Bucket

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Maximum target size of buckets. Enter 'auto_high_volume' for high-volume indexes.

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Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Review >

Input Settings

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Regular expression on path

Segment in path

Host field value

LAPTOP-EHJ3GFJI

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FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

Index

Default

Create a new index

Default

dns_lab

history

main

summary

127.0.0.1:8000/en-US/manager/search/adddatamethods/inputsettings#

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Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< BackReview>

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Indexdns_labCreate a new index

FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

splunk>enterprise

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Add Data

Select SourceSet Source TypeInput SettingsReviewDone

< BackSubmit>

Review

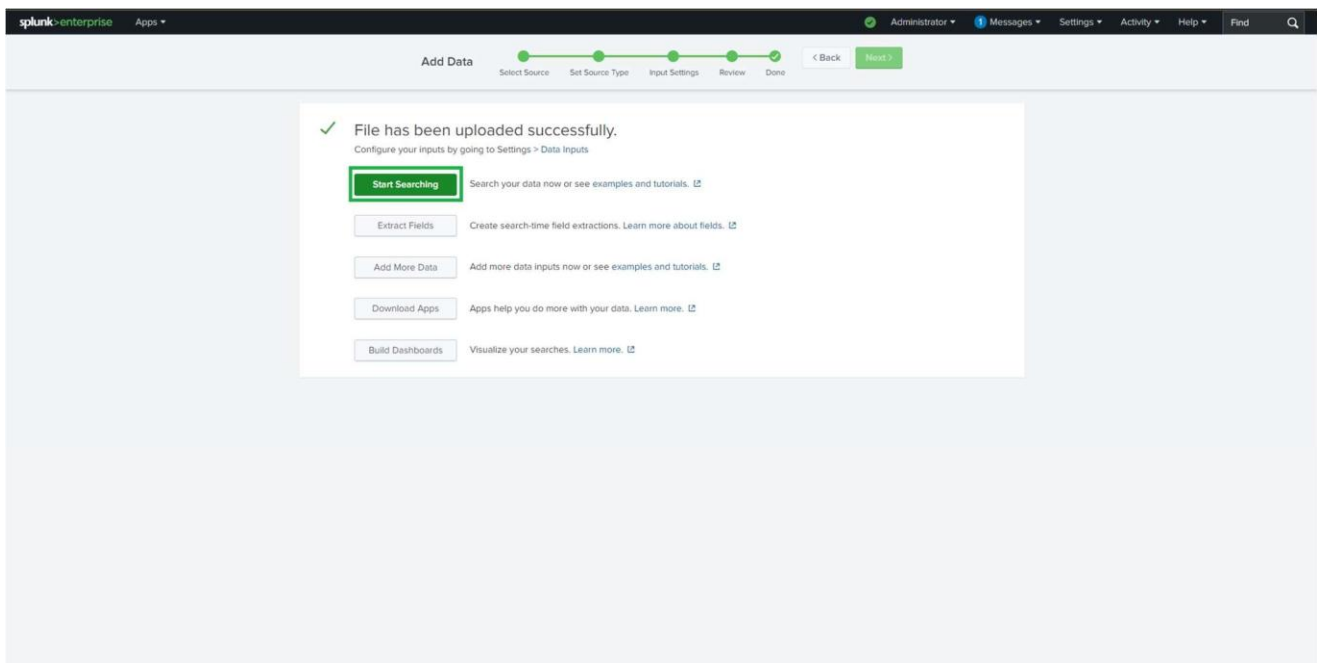
Input Type..... Uploaded File

File Name..... dns_logs.json

Source Type..... json

Host..... LAPTOP-EHJ3QFJI

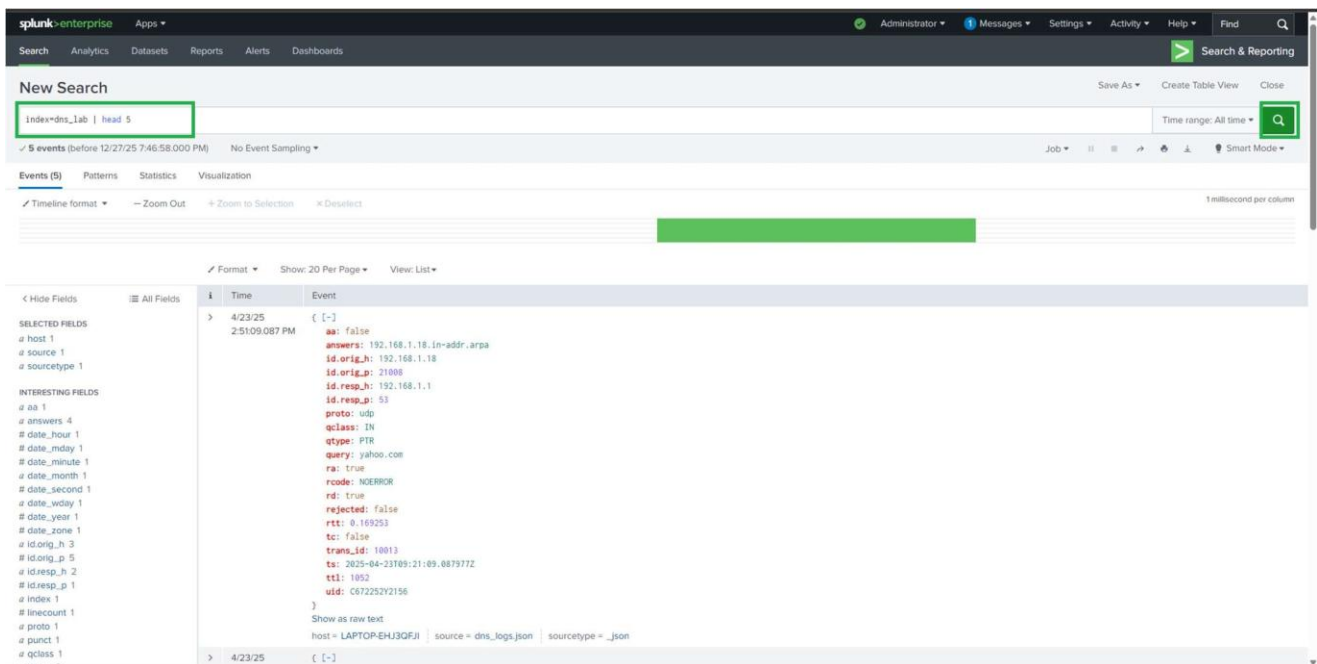
Index..... dns_lab



Step 2: Verify Data Ingestion

Run the following query to confirm logs are indexed: index=dns_lab

| head 5



Lab Tasks & SPL Queries

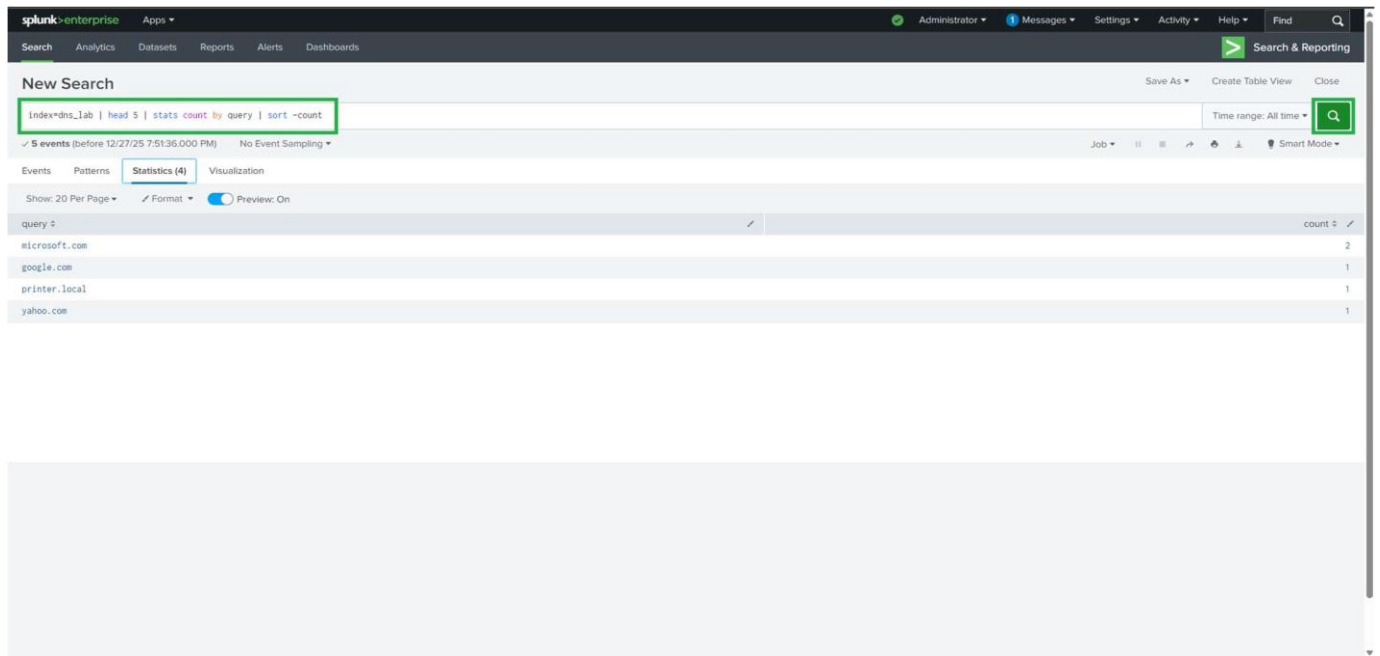
Task 1: Most Frequently Queried Domain Names

Identify which domains are queried most often.

```
index=dns_lab | stats count by query
```

```
| sort -count
```

Use Case: Detect suspicious domains, beaconing behaviour, or malware C2 lookups.



The screenshot shows the Splunk Enterprise Search interface. The search bar contains the query: `index=dns_lab | head 5 | stats count by query | sort -count`. The search results are displayed in a table with the following data:

query	count
microsoft.com	2
google.com	1
printer.local	1
yahoo.com	1

Task 2: Most Active Client IPs

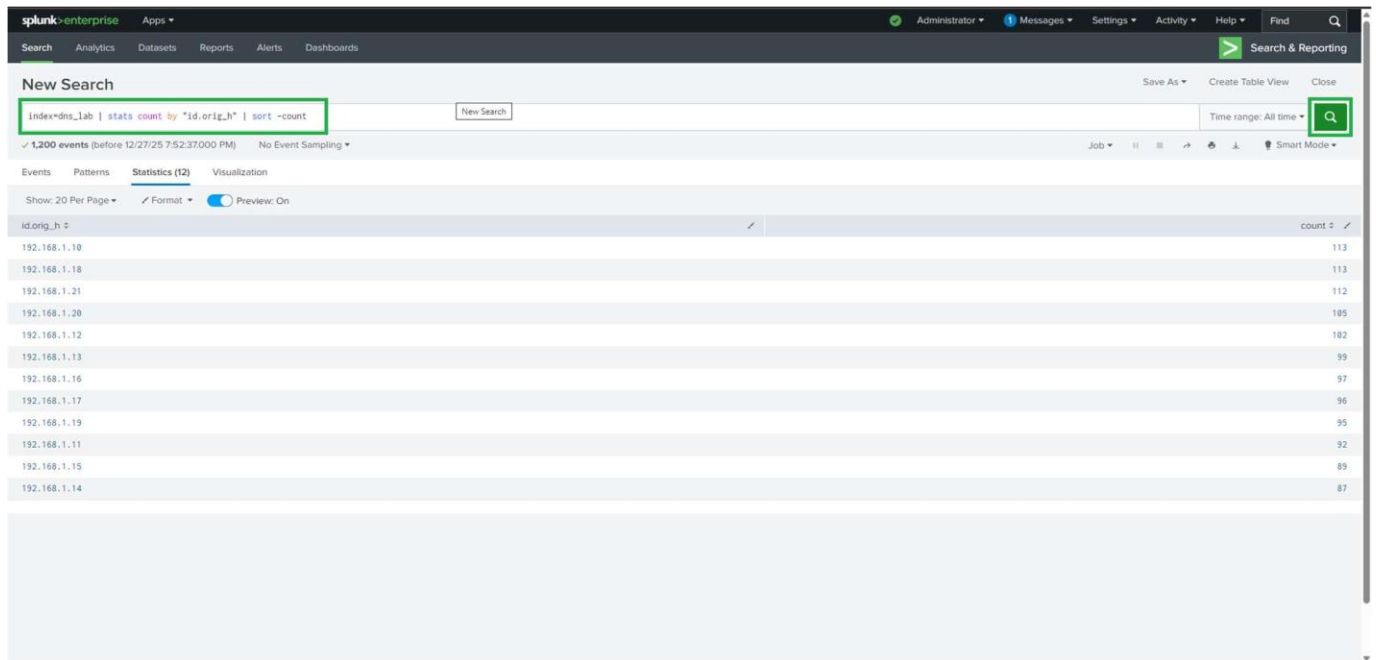
Find which internal hosts generate the most DNS traffic.

index=dns_lab

| stats count by "id.orig_h"

| sort -count

Use Case: Identify compromised systems or misconfigured devices.



The screenshot shows the Splunk Enterprise Search & Reporting interface. The search bar contains the query: `index=dns_lab | stats count by "id.orig_h" | sort -count`. The search results show 1,200 events. The table displays the top 12 results, sorted by count in descending order.

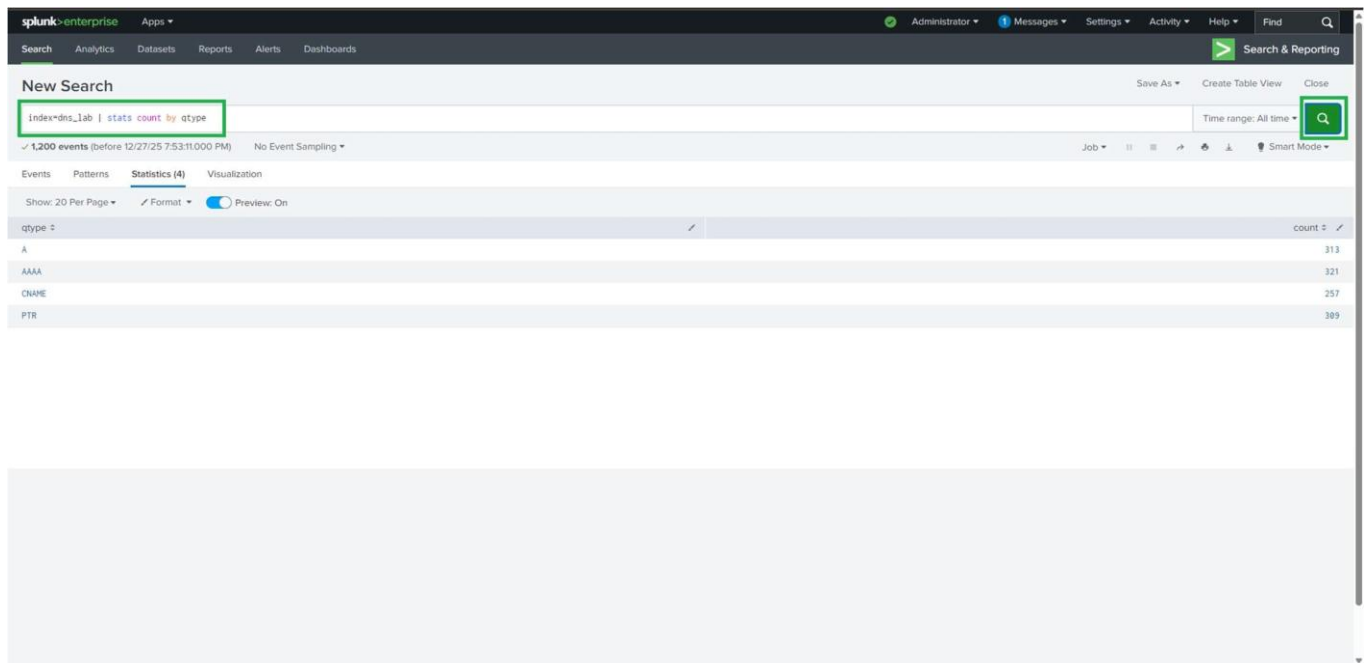
id.orig_h	count
192.168.1.10	113
192.168.1.18	113
192.168.1.21	112
192.168.1.20	105
192.168.1.12	102
192.168.1.13	99
192.168.1.16	97
192.168.1.17	96
192.168.1.19	95
192.168.1.11	92
192.168.1.15	89
192.168.1.14	87

Analyze DNS record types used in the environment. index=dns_lab

| stats count by qtype Use

Case:

- PTR → Reverse lookups
- Excessive AAAA → IPv6 behavior
- CNAME → Redirect or alias usage



The screenshot shows the Splunk Enterprise Search interface. The search bar contains the query `index=dns_lab | stats count by qtype`. The search results are displayed in a table with the following data:

qtype	count
A	313
AAAA	321
CNAME	257
PTR	309

Key Findings & Analysis

- Identified top queried domains such as Google, Microsoft, and Yahoo
- Detected high-volume DNS-generating hosts
- Observed distribution of A, AAAA, CNAME, and PTR records
- Analyzed response times (RTT) and failed lookups (NXDOMAIN)
- Established baseline DNS behavior for anomaly detection

Security Insights

This project enables detection of:

- DNS tunneling patterns

- Malware beaconing via repeated queries
- Suspicious reverse DNS lookups
- Unusual DNS latency or failed resolutions

Dashboards & Alerts (Optional Enhancements) You

can extend this project by:

- Creating Splunk dashboards for DNS activity visualization  Setting alerts for:
 - High DNS query volume
 - Rare domain lookups
 - Excessive failed DNS responses

Conclusion

By completing this lab, you have:

- ✓ Successfully ingested and parsed DNS logs in Splunk
- ✓ Built SPL queries for DNS traffic analysis
- ✓ Identified top domains and active clients
- ✓ Analyzed DNS record types
- ✓ Gained hands-on experience in security-focused log analysis

This project serves as a strong foundation for SOC, SIEM, and threat-hunting roles.
